

Matrices & Vectors

ANSWER KEY



Addition and scalar multiplication

- 1 Let A be the matrix with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and B be the matrix with rows $\langle -1, 4 \rangle$ and $\langle 0, 2 \rangle$. Find $A + B$ and $3A$.

$A + B$ has rows $\langle 1, 5 \rangle$ and $\langle 3, 1 \rangle$. $3A$ has rows $\langle 6, 3 \rangle$ and $\langle 9, -3 \rangle$.

- 2 Using the same A and B , find $A - 2B$.

$A - 2B$ has rows $\langle 2 - (-2), 1 - 8 \rangle = \langle 4, -7 \rangle$ and $\langle 3 - 0, -1 - 4 \rangle = \langle 3, -5 \rangle$.

Matrix multiplication

- 3 Using A (rows $\langle 2, 1 \rangle, \langle 3, -1 \rangle$) and B (rows $\langle -1, 4 \rangle, \langle 0, 2 \rangle$) from above, find AB .

AB has rows $\langle -2, 10 \rangle$ and $\langle -3, 10 \rangle$: e.g. row 1, column 1 is $(2)(-1) + (1)(0) = -2$.

- 4 Let A have rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and let $\mathbf{v} = \langle 2, -3 \rangle$. Find $A\mathbf{v}$.

$A\mathbf{v} = \langle (2)(2) + (1)(-3), (3)(2) + (-1)(-3) \rangle = \langle 1, 9 \rangle$.

- 5 Explain, without computing BA , why AB and BA need not be equal for 2×2 matrices.

Matrix multiplication combines rows of the first factor with columns of the second; swapping the order pairs different rows with different columns, so in general $AB \neq BA$ (matrix multiplication is not commutative).

Determinant and inverse

- 6 Find $\det A$ for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$, and state whether A is invertible.

$\det A = (2)(-1) - (1)(3) = -5 \neq 0$, so A is invertible.

- 7 Find A^{-1} for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, -1 \rangle$.

$A^{-1} = \frac{1}{-5} \times$ the matrix with rows $\langle -1, -1 \rangle$ and $\langle -3, 2 \rangle$, i.e., A^{-1} has rows $\langle 0.2, 0.2 \rangle$ and $\langle 0.6, -0.4 \rangle$.

- 8 Without computing it, explain why the matrix with rows $\langle 4, 2 \rangle$ and $\langle 6, 3 \rangle$ has no inverse.

$\det = (4)(3) - (2)(6) = 12 - 12 = 0$, and a matrix with determinant 0 has no inverse (Theorem 2.1).

Solving systems with the inverse matrix

- 9 Solve the system $3x - y = 8$ and $x + 2y = 5$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

M has rows $\langle 3, -1 \rangle$ and $\langle 1, 2 \rangle$, so $\det M = (3)(2) - (-1)(1) = 7$, and M^{-1} has rows $\langle \frac{2}{7}, \frac{1}{7} \rangle$ and $\langle -\frac{1}{7}, \frac{3}{7} \rangle$. Then $x = \frac{1}{7}[2(8) + 1(5)] = 3$ and $y = \frac{1}{7}[-1(8) + 3(5)] = 1$. Check: $3(3) - 1 = 8$ and $3 + 2(1) = 5$.

Vector magnitude and dot product

- 10 For $\mathbf{u} = \langle 3, 4 \rangle$ and $\mathbf{w} = \langle -4, 3 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

$$\mathbf{u} \cdot \mathbf{w} = (3)(-4) + (4)(3) = 0, \text{ and } |\mathbf{u}| = \sqrt{3^2 + 4^2} = 5.$$

- 11 What does $\mathbf{u} \cdot \mathbf{w} = 0$ tell you about $\mathbf{u}$ and $\mathbf{w}$ from the previous question?

A zero dot product means $\mathbf{u}$ and $\mathbf{w}$ are perpendicular (orthogonal).

Angle between vectors

- 12 Find the angle between $\mathbf{p} = \langle 1, 0 \rangle$ and $\mathbf{q} = \langle 1, 1 \rangle$.

$$\mathbf{p} \cdot \mathbf{q} = 1, |\mathbf{p}| = 1, |\mathbf{q}| = \sqrt{2}. \cos \theta = \frac{1}{\sqrt{2}} \approx 0.7071 \Rightarrow \theta = 45^\circ.$$

- 13 Find the angle between $\langle 6, -8 \rangle$ and $\langle 3, 4 \rangle$, rounded to the nearest tenth of a degree.

$$\text{Dot product} = 18 - 32 = -14, \text{ magnitudes } 10 \text{ and } 5. \cos \theta = \frac{-14}{50} = -0.28 \Rightarrow \theta \approx 106.3^\circ.$$

Multiple choice — easy

- 14 Find $\det A$ for the matrix A with rows $\langle -2, 9 \rangle$ and $\langle -5, -1 \rangle$.

- a. 19
- b. 47 ✓
- c. -43
- d. 49

$$\det A = (-2)(-1) - (9)(-5) = 2 + 45 = 47.$$

15 Find $\det A$ for the matrix A with rows $\langle 4, 8 \rangle$ and $\langle -4, 6 \rangle$.

- a. -64
- b. 58
- c. 56 ✓
- d. -8

$$\det A = (4)(6) - (8)(-4) = 24 + 32 = 56.$$

16 Find $\det A$ for the matrix A with rows $\langle 2, -2 \rangle$ and $\langle 9, -8 \rangle$.

- a. 5
- b. -34
- c. 2 ✓
- d. 4

$$\det A = (2)(-8) - (-2)(9) = -16 + 18 = 2.$$

17 Find $\det A$ for the matrix A with rows $\langle 0, -4 \rangle$ and $\langle 1, 4 \rangle$.

- a. -4
- b. 4 ✓
- c. 16
- d. 6

$$\det A = (0)(4) - (-4)(1) = 0 + 4 = 4.$$

18 Find $\det A$ for the matrix A with rows $\langle -4, -8 \rangle$ and $\langle -1, -3 \rangle$.

- a. 20
- b. -20
- c. 4 ✓
- d. 6

$$\det A = (-4)(-3) - (-8)(-1) = 12 - 8 = 4.$$

19 Find $\det A$ for the matrix A with rows $\langle -1, -4 \rangle$ and $\langle 5, -1 \rangle$.

a. -19

b. 23

c. -9

d. 21 ✓

$$\det A = (-1)(-1) - (-4)(5) = 1 + 20 = 21.$$

20 Find $\det A$ for the matrix A with rows $\langle 0, -8 \rangle$ and $\langle -4, 4 \rangle$.

a. -30

b. 32

c. -29

d. -32 ✓

$$\det A = (0)(4) - (-8)(-4) = 0 - 32 = -32.$$

21 Find $\det A$ for the matrix A with rows $\langle -5, 3 \rangle$ and $\langle -8, 6 \rangle$.

a. -4

b. -54

c. 22

d. -6 ✓

$$\det A = (-5)(6) - (3)(-8) = -30 + 24 = -6.$$

22 Find $\det A$ for the matrix A with rows $\langle -5, 0 \rangle$ and $\langle -7, 8 \rangle$.

a. -40 ✓

b. 35

c. -37

d. -38

$$\det A = (-5)(8) - (0)(-7) = -40 = -40.$$

23 Find $\det A$ for the matrix A with rows $\langle -1, -3 \rangle$ and $\langle -9, 9 \rangle$.

a. -34

b. -36 ✓

c. 36

d. 18

$$\det A = (-1)(9) - (-3)(-9) = -9 - 27 = -36.$$

24 Find $\det A$ for the matrix A with rows $\langle 7, -5 \rangle$ and $\langle 6, -4 \rangle$.

a. 2 ✓

b. 4

c. -58

d. 22

$$\det A = (7)(-4) - (-5)(6) = -28 + 30 = 2.$$

25 Find $\det A$ for the matrix A with rows $\langle -2, 9 \rangle$ and $\langle -7, -1 \rangle$.

a. 67

b. -61

c. 65 ✓

d. 23

$$\det A = (-2)(-1) - (9)(-7) = 2 + 63 = 65.$$

26 Find $\det A$ for the matrix A with rows $\langle -7, 9 \rangle$ and $\langle 6, -7 \rangle$.

a. -5 ✓

b. 103

c. -3

d. 21

$$\det A = (-7)(-7) - (9)(6) = 49 - 54 = -5.$$

27 Find $\det A$ for the matrix A with rows $\langle -1, -5 \rangle$ and $\langle 2, -3 \rangle$.

a. -7

b. 15

c. -17

d. 13 ✓

$$\det A = (-1)(-3) - (-5)(2) = 3 + 10 = 13.$$

28 Find $\det A$ for the matrix A with rows $\langle 5, 2 \rangle$ and $\langle -9, -3 \rangle$.

a. -33

b. 3 ✓

c. 5

d. -39

$$\det A = (5)(-3) - (2)(-9) = -15 + 18 = 3.$$

Multiple choice — medium

29 Let A have rows $\langle 4, 2 \rangle$ and $\langle 3, 7 \rangle$, and let $\mathbf{v} = \langle 5, 2 \rangle$. Find $A\mathbf{v}$.

- a. $\langle 26, 29 \rangle$
- b. $\langle 24, 27 \rangle$
- c. $\langle 29, 24 \rangle$
- d. $\langle 24, 29 \rangle$ ✓

$$A\mathbf{v} = \langle (4)(5) + (2)(2), (3)(5) + (7)(2) \rangle = \langle 24, 29 \rangle.$$

30 Let A have rows $\langle -3, 7 \rangle$ and $\langle 1, 3 \rangle$, and let $\mathbf{v} = \langle 3, -5 \rangle$. Find $A\mathbf{v}$.

- a. $\langle -42, -12 \rangle$
- b. $\langle -12, -44 \rangle$
- c. $\langle -44, -12 \rangle$ ✓
- d. $\langle -44, -14 \rangle$

$$A\mathbf{v} = \langle (-3)(3) + (7)(-5), (1)(3) + (3)(-5) \rangle = \langle -44, -12 \rangle.$$

31 Let A have rows $\langle -4, 4 \rangle$ and $\langle 8, 4 \rangle$, and let $\mathbf{v} = \langle 5, 3 \rangle$. Find $A\mathbf{v}$.

- a. $\langle -6, 52 \rangle$
- b. $\langle 52, -8 \rangle$
- c. $\langle -8, 50 \rangle$
- d. $\langle -8, 52 \rangle$ ✓

$$A\mathbf{v} = \langle (-4)(5) + (4)(3), (8)(5) + (4)(3) \rangle = \langle -8, 52 \rangle.$$

32 Let A have rows $\langle 1, 1 \rangle$ and $\langle 7, -7 \rangle$, and let $\mathbf{v} = \langle 7, 7 \rangle$. Find $A\mathbf{v}$.

- a. $\langle 14, 0 \rangle$ ✓
- b. $\langle 0, 14 \rangle$
- c. $\langle 14, -2 \rangle$
- d. $\langle 16, 0 \rangle$

$$A\mathbf{v} = \langle (1)(7) + (1)(7), (7)(7) + (-7)(7) \rangle = \langle 14, 0 \rangle.$$

33 Let A have rows $\langle 9, -6 \rangle$ and $\langle -4, 3 \rangle$, and let $\mathbf{v} = \langle -2, -5 \rangle$. Find $A\mathbf{v}$.

a. $\langle 12, -9 \rangle$

b. $\langle -7, 12 \rangle$

c. $\langle 14, -7 \rangle$

d. $\langle 12, -7 \rangle$ ✓

$$A\mathbf{v} = \langle (9)(-2) + (-6)(-5), (-4)(-2) + (3)(-5) \rangle = \langle 12, -7 \rangle.$$

34 Let A have rows $\langle -1, 6 \rangle$ and $\langle 2, 2 \rangle$, and let $\mathbf{v} = \langle -1, 5 \rangle$. Find $A\mathbf{v}$.

a. $\langle 33, 8 \rangle$

b. $\langle 31, 8 \rangle$ ✓

c. $\langle 8, 31 \rangle$

d. $\langle 31, 6 \rangle$

$$A\mathbf{v} = \langle (-1)(-1) + (6)(5), (2)(-1) + (2)(5) \rangle = \langle 31, 8 \rangle.$$

35 Let A have rows $\langle 9, -9 \rangle$ and $\langle -1, 7 \rangle$, and let $\mathbf{v} = \langle -8, -8 \rangle$. Find $A\mathbf{v}$.

a. $\langle 2, -48 \rangle$

b. $\langle 0, -48 \rangle$ ✓

c. $\langle 0, -50 \rangle$

d. $\langle -48, 0 \rangle$

$$A\mathbf{v} = \langle (9)(-8) + (-9)(-8), (-1)(-8) + (7)(-8) \rangle = \langle 0, -48 \rangle.$$

36 Let A have rows $\langle -7, -6 \rangle$ and $\langle 8, 4 \rangle$, and let $\mathbf{v} = \langle -5, -4 \rangle$. Find $A\mathbf{v}$.

a. $\langle 61, -56 \rangle$

b. $\langle 59, -56 \rangle$ ✓

c. $\langle 59, -58 \rangle$

d. $\langle -56, 59 \rangle$

$$A\mathbf{v} = \langle (-7)(-5) + (-6)(-4), (8)(-5) + (4)(-4) \rangle = \langle 59, -56 \rangle.$$

37 Let A have rows $\langle -9, -8 \rangle$ and $\langle 2, -4 \rangle$, and let $\mathbf{v} = \langle 9, -1 \rangle$. Find $A\mathbf{v}$.

a. $\langle -73, 22 \rangle$ ✓

b. $\langle 22, -73 \rangle$

c. $\langle -71, 22 \rangle$

d. $\langle -73, 20 \rangle$

$$A\mathbf{v} = \langle (-9)(9) + (-8)(-1), (2)(9) + (-4)(-1) \rangle = \langle -73, 22 \rangle.$$

38 Let A have rows $\langle -2, -6 \rangle$ and $\langle 7, -5 \rangle$, and let $\mathbf{v} = \langle -9, -3 \rangle$. Find $A\mathbf{v}$.

a. $\langle 38, -48 \rangle$

b. $\langle 36, -48 \rangle$ ✓

c. $\langle 36, -50 \rangle$

d. $\langle -48, 36 \rangle$

$$A\mathbf{v} = \langle (-2)(-9) + (-6)(-3), (7)(-9) + (-5)(-3) \rangle = \langle 36, -48 \rangle.$$

39 Let A have rows $\langle -8, -1 \rangle$ and $\langle -7, -9 \rangle$, and let $\mathbf{v} = \langle 9, -9 \rangle$. Find $A\mathbf{v}$.

a. $\langle -63, 16 \rangle$

b. $\langle -61, 18 \rangle$

c. $\langle 18, -63 \rangle$

d. $\langle -63, 18 \rangle$ ✓

$$A\mathbf{v} = \langle (-8)(9) + (-1)(-9), (-7)(9) + (-9)(-9) \rangle = \langle -63, 18 \rangle.$$

40 Let A have rows $\langle -8, -4 \rangle$ and $\langle 3, 6 \rangle$, and let $\mathbf{v} = \langle -4, -5 \rangle$. Find $A\mathbf{v}$.

a. $\langle -42, 52 \rangle$

b. $\langle 52, -44 \rangle$

c. $\langle 54, -42 \rangle$

d. $\langle 52, -42 \rangle$ ✓

$$A\mathbf{v} = \langle (-8)(-4) + (-4)(-5), (3)(-4) + (6)(-5) \rangle = \langle 52, -42 \rangle.$$

41 Let A have rows $\langle -7, -2 \rangle$ and $\langle 1, 0 \rangle$, and let $\mathbf{v} = \langle 8, -1 \rangle$. Find $A\mathbf{v}$.

a. $\langle -54, 6 \rangle$

b. $\langle 8, -54 \rangle$

c. $\langle -52, 8 \rangle$

d. $\langle -54, 8 \rangle$ ✓

$$A\mathbf{v} = \langle (-7)(8) + (-2)(-1), (1)(8) + (0)(-1) \rangle = \langle -54, 8 \rangle.$$

42 Let A have rows $\langle -6, -4 \rangle$ and $\langle -5, -2 \rangle$, and let $\mathbf{v} = \langle 7, 9 \rangle$. Find $A\mathbf{v}$.

a. $\langle -76, -53 \rangle$

b. $\langle -78, -53 \rangle$ ✓

c. $\langle -78, -55 \rangle$

d. $\langle -53, -78 \rangle$

$$A\mathbf{v} = \langle (-6)(7) + (-4)(9), (-5)(7) + (-2)(9) \rangle = \langle -78, -53 \rangle.$$

43 Let A have rows $\langle 3, -2 \rangle$ and $\langle -5, 1 \rangle$, and let $\mathbf{v} = \langle 5, 5 \rangle$. Find $A\mathbf{v}$.

a. $\langle 5, -22 \rangle$

b. $\langle 7, -20 \rangle$

c. $\langle 5, -20 \rangle$ ✓

d. $\langle -20, 5 \rangle$

$$A\mathbf{v} = \langle (3)(5) + (-2)(5), (-5)(5) + (1)(5) \rangle = \langle 5, -20 \rangle.$$

Multiple choice — hard

44 Find A^{-1} for the matrix A with rows $\langle 4, -5 \rangle$ and $\langle -5, 6 \rangle$.

a. $\langle -6, -5 \rangle, \langle -5, -4 \rangle$ ✓

b. $\langle 4, -5 \rangle, \langle -5, 6 \rangle$

c. $\langle -5, -5 \rangle, \langle -5, -4 \rangle$

d. $\langle -5, -4 \rangle, \langle -6, -5 \rangle$

$$\det A = (4)(6) - (-5)(-5) = -1. A^{-1} = \frac{1}{-1} \times \text{the matrix with rows } \langle 6, 5 \rangle \text{ and } \langle 5, 4 \rangle, \text{ giving rows } \langle -6, -5 \rangle \text{ and } \langle -5, -4 \rangle.$$

45 Find A^{-1} for the matrix A with rows $\langle 3, 1 \rangle$ and $\langle -1, 0 \rangle$.

a. $\langle 0, -1 \rangle, \langle 1, 3 \rangle$ ✓

b. $\langle 0, 1 \rangle, \langle -1, 3 \rangle$

c. $\langle 1, 3 \rangle, \langle 0, -1 \rangle$

d. $\langle 3, 1 \rangle, \langle -1, 0 \rangle$

$$\det A = (3)(0) - (1)(-1) = 1. A^{-1} = \frac{1}{1} \times \text{the matrix with rows } \langle 0, -1 \rangle \text{ and } \langle 1, 3 \rangle, \text{ giving rows } \langle 0, -1 \rangle \text{ and } \langle 1, 3 \rangle.$$

46 Find A^{-1} for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle 3, 1 \rangle$.

a. $\langle -1, 3 \rangle, \langle 1, -2 \rangle$

b. $\langle 2, 1 \rangle, \langle 3, 1 \rangle$

c. $\langle -1, 1 \rangle, \langle 3, -2 \rangle$ ✓

d. $\langle 3, -2 \rangle, \langle -1, 1 \rangle$

$\det A = (2)(1) - (1)(3) = -1$. $A^{-1} = \frac{1}{-1} \times$ the matrix with rows $\langle 1, -1 \rangle$ and $\langle -3, 2 \rangle$, giving rows $\langle -1, 1 \rangle$ and $\langle 3, -2 \rangle$.

47 Find A^{-1} for the matrix A with rows $\langle 1, -2 \rangle$ and $\langle 2, -5 \rangle$.

a. $\langle 5, 2 \rangle, \langle -2, -1 \rangle$

b. $\langle 2, -1 \rangle, \langle 5, -2 \rangle$

c. $\langle 5, -2 \rangle, \langle 2, -1 \rangle$ ✓

d. $\langle 1, -2 \rangle, \langle 2, -5 \rangle$

$\det A = (1)(-5) - (-2)(2) = -1$. $A^{-1} = \frac{1}{-1} \times$ the matrix with rows $\langle -5, 2 \rangle$ and $\langle -2, 1 \rangle$, giving rows $\langle 5, -2 \rangle$ and $\langle 2, -1 \rangle$.

48 Find A^{-1} for the matrix A with rows $\langle 2, 1 \rangle$ and $\langle -5, -3 \rangle$.

a. $\langle 3, -5 \rangle, \langle 1, -2 \rangle$

b. $\langle 2, 1 \rangle, \langle -5, -3 \rangle$

c. $\langle 3, 1 \rangle, \langle -5, -2 \rangle$ ✓

d. $\langle -5, -2 \rangle, \langle 3, 1 \rangle$

$\det A = (2)(-3) - (1)(-5) = -1$. $A^{-1} = \frac{1}{-1} \times$ the matrix with rows $\langle -3, -1 \rangle$ and $\langle 5, 2 \rangle$, giving rows $\langle 3, 1 \rangle$ and $\langle -5, -2 \rangle$.

49 Solve the system $6y = -30$ and $6x = 24$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle -5, 4 \rangle$

b. $\langle 6, -5 \rangle$

c. $\langle 4, -5 \rangle$ ✓

d. $\langle 4, -7 \rangle$

M has rows $(0, 6)$ and $(6, 0)$, so $\det M = (0)(0) - (6)(6) = -36$. Solving gives $x = 4$ and $y = -5$.
Check: $(0)(4) - 30 = -30$ and $(6)(4) = 24$.

50 Solve the system $4x - 5y = -3$ and $-6x + 4y = -6$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle 3, 1 \rangle$

b. $\langle 3, 3 \rangle$ ✓

c. $\langle 6, 3 \rangle$

d. $\langle 5, 3 \rangle$

M has rows $(4, -5)$ and $(-6, 4)$, so $\det M = (4)(4) - (-5)(-6) = -14$. Solving gives $x = 3$ and $y = 3$.
Check: $(4)(3) - 15 = -3$ and $(-6)(3) + 12 = -6$.

51 Solve the system $x + 3y = -15$ and $x + 2y = -8$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle 6, -7 \rangle$ ✓

b. $\langle -7, 6 \rangle$

c. $\langle 8, -7 \rangle$

d. $\langle 6, -9 \rangle$

M has rows $(1, 3)$ and $(1, 2)$, so $\det M = (1)(2) - (3)(1) = -1$. Solving gives $x = 6$ and $y = -7$. Check:
 $(1)(6) - 21 = -15$ and $(1)(6) - 14 = -8$.

52 Solve the system $2x - y = 5$ and $6x + 6y = -48$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle -1, -7 \rangle$ ✓

b. $\langle 1, -7 \rangle$

c. $\langle -1, -9 \rangle$

d. $\langle -7, -1 \rangle$

M has rows $\langle 2, -1 \rangle$ and $\langle 6, 6 \rangle$, so $\det M = (2)(6) - (-1)(6) = 18$. Solving gives $x = -1$ and $y = -7$.
Check: $(2)(-1) + 7 = 5$ and $(6)(-1) - 42 = -48$.

53 Solve the system $x - y = 0$ and $-3x - 6y = -54$ using $\mathbf{x} = M^{-1}\mathbf{b}$.

a. $\langle 9, 6 \rangle$

b. $\langle 6, 4 \rangle$

c. $\langle 6, 6 \rangle$ ✓

d. $\langle 8, 6 \rangle$

M has rows $\langle 1, -1 \rangle$ and $\langle -3, -6 \rangle$, so $\det M = (1)(-6) - (-1)(-3) = -9$. Solving gives $x = 6$ and $y = 6$. Check: $(1)(6) - 6 = 0$ and $(-3)(6) - 36 = -54$.

54 For $\mathbf{u} = \langle 20, 21 \rangle$ and $\mathbf{v} = \langle -5, -12 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{352}{377}$

b. $-\frac{352}{377}$ ✓

c. $-\frac{27}{29}$

d. $-\frac{176}{21}$

$\mathbf{u} \cdot \mathbf{v} = (20)(-5) + (21)(-12) = -352$. $|\mathbf{u}| = 29$, $|\mathbf{v}| = 13$. $\cos \theta = -352/377 = -\frac{352}{377}$.

- 55 For $\mathbf{u} = \langle -20, -21 \rangle$ and $\mathbf{v} = \langle 8, -15 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{155}{493}$ ✓

b. $-\frac{155}{493}$

c. $\frac{156}{493}$

d. $\frac{155}{46}$

$$\mathbf{u} \cdot \mathbf{v} = (-20)(8) + (-21)(-15) = 155. |\mathbf{u}| = 29, |\mathbf{v}| = 17. \cos \theta = 155/493 = \frac{155}{493}.$$

- 56 For $\mathbf{u} = \langle 20, -21 \rangle$ and $\mathbf{v} = \langle -5, -12 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{152}{377}$ ✓

b. $\frac{76}{21}$

c. $\frac{153}{377}$

d. $-\frac{152}{377}$

$$\mathbf{u} \cdot \mathbf{v} = (20)(-5) + (-21)(-12) = 152. |\mathbf{u}| = 29, |\mathbf{v}| = 13. \cos \theta = 152/377 = \frac{152}{377}.$$

- 57 For $\mathbf{u} = \langle 8, 15 \rangle$ and $\mathbf{v} = \langle 7, 24 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{417}{425}$

b. $\frac{208}{21}$

c. $-\frac{416}{425}$

d. $\frac{416}{425}$ ✓

$$\mathbf{u} \cdot \mathbf{v} = (8)(7) + (15)(24) = 416. |\mathbf{u}| = 17, |\mathbf{v}| = 25. \cos \theta = 416/425 = \frac{416}{425}.$$

58 For $\mathbf{u} = \langle 7, -24 \rangle$ and $\mathbf{v} = \langle 5, 12 \rangle$, find $\cos \theta$ where θ is the angle between them, using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.

a. $\frac{253}{325}$

b. $-\frac{253}{38}$

c. $-\frac{252}{325}$

d. $-\frac{253}{325}$ ✓

$$\mathbf{u} \cdot \mathbf{v} = (7)(5) + (-24)(12) = -253. |\mathbf{u}| = 25, |\mathbf{v}| = 13. \cos \theta = -253/325 = -\frac{253}{325}.$$